SQL_Project

● Analyzing sales using salesman table and order table

● Customer segmentation using customer table (i.e. grouping customer using purchase history).

● Study with data using table such as customer, salesman and order table.

● Analyze nobel_win table.

Salesman Table

Query :-

```
CREATE TABLE salesman(
salesman_id number primary key,
name varchar(20),
city varchar(20),
commission float(5)
);

insert into salesman VALUES (5001, 'James Hoog', 'New York', 0.15);
insert into salesman VALUES (5002, 'Nail Knite', 'Paris', 0.13);
insert into salesman VALUES (5005, 'Pit Alex', 'London', 0.11);
insert into salesman VALUES (5006, 'Mc Lyon', 'Paris', 0.14);
insert into salesman VALUES (5003, 'Lauson Hen','', 0.12);
insert into salesman VALUES (5007, 'Paul Adam', 'Rome', 0.13);
```

Output :-

	SALESMAN_ID	NAME	CITY	COMMISSION
1	5001	James Hoog	New York	0.15
2	5002	Nail Knite	Paris	0.13
3	5005	Pit Alex	London	0.11
4	5006	Mc Lyon	Paris	0.14
5	5003	Lauson Hen	(null)	0.12
6	5007	Paul Adam	Rome	0.13

Customer Table

Query :-

```
CREATE TABLE customer(
customer_id number primary key,
customer_name varchar(20),
customer_city varchar(20),
grade number(5),
salesman_id number(5),
foreign key (salesman_id) references salesman(salesman_id)
);

insert into customer values (3002, 'Nick Rimando', 'New York', 100, 5001);
insert into customer values (3005, 'Graham Zusi', 'California', 200, 5002);
insert into customer values (3001, 'Brad Guzan', 'London', '', '');
insert into customer values (3004, 'Fabian Johns', 'Paris', 300, 5006);
insert into customer values (3007, 'Brad Davis', 'New York', 200, 5001);
insert into customer values (3009, 'Geoff Cameo', 'Berlin', 100, '');
insert into customer values (3008, 'Julian Green', 'London', 300, 5002);
insert into customer values (3003, 'Jozy Altidor', 'Moncow', 200, 5007);
```

Output :-

	CUSTOMER_ID	CUSTOMER_NAME	CUSTOMER_CITY	GRADE	SALESMAN_ID
1	3002	Nick Rimando	New York	100	5001
2	3005	Graham Zusi	California	200	5002
3	3001	Brad Guzan	London	(null)	(null)
4	3004	Fabian Johns	Paris	300	5006
5	3007	Brad Davis	New York	200	5001
6	3009	Geoff Cameo	Berlin	100	(null)
7	3008	Julian Green	London	300	5002
8	3003	Jozy Altidor	Moncow	200	5007

Orders Table

Query :-

```
CREATE TABLE orders(
order_no number primary key,
punch_amt float(10),
```

```
order_date date,
customer_id number(5),
salesman_id number(5),
foreign key (salesman_id) references salesman(salesman_id),
foreign key (customer_id) references customer(customer_id)
);

insert into orders values (7001, 150.5, to_date('05-OCT-16', 'dd-mon-yy'), 3005, 5002);
insert into orders values (7009, 270.65, to_date('10-SEP-16', 'dd-mon-yy'), 3001, '');
insert into orders values (7002, 65.26, to_date('05-OCT-16', 'dd-mon-yy'), 3002, 5001);
insert into orders values (7004, 110.5, to_date('17-AUG-16', 'dd-mon-yy'), 3009, '');
insert into orders values (7007, 948.5, to_date('10-SEP-16', 'dd-mon-yy'), 3005, 5002);
insert into orders values (7005, 2400.6, to_date('27-JUL-16', 'dd-mon-yy'), 3007, 5001);
insert into orders values (7008, 5760, to_date('10-SEP-16', 'dd-mon-yy'), 3002, 5001);
insert into orders values (7010, 1983.43, to_date('10-OCT-16', 'dd-mon-yy'), 3004, 5006);
insert into orders values (7003, 2480.4, to_date('10-OCT-16', 'dd-mon-yy'), 3009, '');
insert into orders values (7012, 250.45, to_date('27-JUN-16', 'dd-mon-yy'), 3008, 5002);
insert into orders values (7011, 75.29, to_date('17-AUG-16', 'dd-mon-yy'), 3003, 5007);
insert into orders values (7013, 75.29, TO_DATE('17-AUG-16', 'DD-MON-YY'), 3003, 5007);
```

Output :-

	ORDER_NO	PUNCH_AMT	ORDER_DATE	CUSTOMER_ID	SALESMAN_ID
1	7001	150.5	05-10-16	3005	5002
2	7009	270.7	10-09-16	3001	(null)
3	7002	65.26	05-10-16	3002	5001
4	7004	110.5	17-08-16	3009	(null)
5	7007	948.5	10-09-16	3005	5002
6	7005	2401	27-07-16	3007	5001
7	7008	5760	10-09-16	3002	5001
8	7010	1983	10-10-16	3004	5006
9	7003	2480	10-10-16	3009	(null)
10	7012	250.5	27-06-16	3008	5002
11	7011	75.29	17-08-16	3003	5007

Nobel_Win Table

Query :-

```
CREATE TABLE Nobel_Win(
year varchar(4),
subject varchar(20),
winner varchar(20),
country varchar(20),
category varchar(20)
);

insert into nobel_win values (1970, 'Physics', 'Hannes Alfven', 'Sweden', 'Scientist');
insert into nobel_win values (1970, 'Physics', 'Louis Neel', 'France', 'Scientist');
insert into nobel_win values (1970, 'Literature', 'Aleksandr solzhenitsyn', 'Russia', 'Linguist');
insert into nobel_win values (1970, 'Chemistry', 'Luis Federico Leloir', 'France', 'Scientist');
insert into nobel_win values (1970, 'Economics', 'Paul Samuelson', 'USA', 'Economist');
insert into nobel_win values (1971, 'Economics', 'Simon Kuznets', 'Russia', 'Economist');
insert into nobel_win values (1971, 'Literature', 'Pablo Neruda', 'Chile', 'Linguist');
insert into nobel_win values (1971, 'Chemistry', 'Gerhard Herzberg', 'Germany', 'Scientist');
insert into nobel_win values (1987, 'Economics', 'Robert Solow', 'USA', 'Economist');
insert into nobel_win values (1987, 'Literature', 'Joseph Brodsky', 'Russia', 'Linguist');
insert into nobel_win values (1987, 'Chemistry', 'Jean-Marie Lehn', 'France', 'Scientist');
insert into nobel_win values (1987, 'Chemistry', 'Donald J. Cram', 'USA', 'Scientist');
insert into nobel_win values (1994, 'Economics', 'Reinhard Selten', 'Germany', 'Economist');
insert into nobel_win values (1994, 'Literature', 'Kenzeburo Oe', 'Japan', 'Linguist');
```

Output :-

	YEAR	SUBJECT	WINNER	COUNTRY	CATEGORY
1	1970	Physics	Hannes Alfven	Sweden	Scientist
2	1970	Physics	Louis Neel	France	Scientist
3	1970	Chemistry	Luis Federico Leloir	France	Scientist
4	1970	Economics	Paul Samuelson	USA	Economist
5	1971	Economics	Simon Kuznets	Russia	Economist
6	1971	Literature	Pablo Neruda	Chile	Linguist
7	1971	Chemistry	Gerhard Herzberg	Germany	Scientist
8	1987	Economics	Robert Solow	USA	Economist
9	1987	Literature	Joseph Brodsky	Russia	Linguist
10	1987	Chemistry	Jean-Marie Lehn	France	Scientist
11	1987	Chemistry	Donald J. Cram	USA	Scientist
12	1994	Economics	Reinhard Selten	Germany	Economist
13	1994	Literature	Kenzeburo Oe	Japan	Linguist

Item_Mast Table

Query :-

```
create table item_mast(
pro_ID number(5),
pro_name varchar(30),
pro_price number(5)
);

insert into item_mast values (101, 'Mother Board', 3200);
insert into item_mast values (102, 'Key Board', 450);
insert into item_mast values (103, 'Zip Drive', 250);
insert into item_mast values (104, 'Speaker', 550);
insert into item_mast values (105, 'Monitor', 5000);
insert into item_mast values (106, 'DVD Drive', 900);
insert into item_mast values (107, 'CD Drive', 800);
insert into item_mast values (108, 'Printer', 2600);
insert into item_mast values (109, 'Cartridge', 350);
insert into item_mast values (110, 'Mouse', 250);
```

Output :-

	PRO_ID	PRO_NAME	PRO_PRICE
1	101	Mother Board	3200
2	102	Key Board	450
3	103	Zip Drive	250
4	104	Speaker	550
5	105	Monitor	5000
6	106	DVD Drive	900
7	107	CD Drive	800
8	108	Printer	2600
9	109	Cartridge	350
10	110	Mouse	250

Query 1 : Display name and commission for all the salesman.

Select name , commission

from salesman;

	NAME	COMMISSION
1	James Hoog	0.15
2	Nail Knite	0.13
3	Pit Alex	0.11
4	Mc Lyon	0.14
5	Lauson Hen	0.12
6	Paul Adam	0.13

Query 2 : Retrieve salesman id of all salesman from orders table without any repeats.

Select distinct salesman_id from orders

where salesman_id is not null;

	SALESMAN_ID
1	5001
2	5006
3	5007
4	5002

Query 3 : Display names and city of salesman, who belongs to the city of Paris.

Select name , city from salesman

where city = 'Paris';

	NAME	CITY
1	Nail Knite	Paris
2	Mc Lyon	Paris

Query 4 : Display all the information for those customers with a grade of 200.

select * from customer

where grade = 200;

	CUSTOMER_ID	CUSTOMER_NAME	CUSTOMER_CITY	GRADE	SALESMAN_ID
1	3005	Graham Zusi	California	200	5002
2	3007	Brad Davis	New York	200	5001
3	3003	Jozy Altidor	Moncow	200	5007

Query 5 : Display the order number, order date and the purchase amount for order(s) which will be delivered by the salesman with ID 5001.

Select order_no , order_date , punch_amt from orders

where salesman_id = 5001;

	ORDER_NO	ORDER_DATE	PUNCH_AMT
1	7002	05-10-16	65.26
2	7005	27-07-16	2401
3	7008	10-09-16	5760

Query 6 : Show the winner of the 1971 prize for Literature.

Select winner from nobel_win

where year = 1971 and subject = 'Literature';

	WINNER
1	Pablo Neruda

Query 7 : Show all the details of the winners with first name Louis.

Select * from nobel_win

where winner like 'Louis%';

	YEAR	SUBJECT	WINNER	COUNTRY	CATEGORY
1	1970	Physics	Louis Neel	France	Scientist

Query 8 : Show all the winners in Physics for 1970 together with the winner of Economics for 1971.

select * from nobel_win

where subject = 'Physics' and year = 1970

union

select * from nobel_win

where subject = 'Economics' and year = 1971;

	YEAR	SUBJECT	WINNER	COUNTRY	CATEGORY
1	1970	Physics	Hannes Alfven	Sweden	Scientist
2	1970	Physics	Louis Neel	France	Scientist
3	1971	Economics	Simon Kuznets	Russia	Economist

Query 9 : Show all the winners of Nobel prize in the year 1970 except the subject Physology and Economics.

```
Select * from nobel_win

where year = 1970 and subject not in ('Physology', 'Economics');
```

	YEAR	SUBJECT	WINNER	COUNTRY	CATEGORY
1	1970	Physics	Hannes Alfven	Sweden	Scientist
2	1970	Physics	Louis Neel	France	Scientist
3	1970	Chemistry	Luis Federico Leloir	France	Scientist

Query 10 : Find all the details of the Nobel winners for the subject not started with the letter 'P' and arranged the list as the most recent comes first, then by name in order.

```
Select * from nobel_win

where subject not like ‘P%’

order by year desc , winner;
```

	YEAR	SUBJECT	WINNER	COUNTRY	CATEGORY
1	1994	Literature	Kenzeburo Oe	Japan	Linguist
2	1994	Economics	Reinhard Selten	Germany	Economist
3	1987	Chemistry	Donald J. Cram	USA	Scientist
4	1987	Chemistry	Jean-Marie Lehn	France	Scientist
5	1987	Literature	Joseph Brodsky	Russia	Linguist
6	1987	Economics	Robert Solow	USA	Economist
7	1971	Chemistry	Gerhard Herzberg	Germany	Scientist
8	1971	Literature	Pablo Neruda	Chile	Linguist
9	1971	Economics	Simon Kuznets	Russia	Economist
10	1970	Chemistry	Luis Federico Leloir	France	Scientist
11	1970	Economics	Paul Samuelson	USA	Economist

Query 11 : Find the name and price of the cheapest item(s).

```
select pro_name,pro_price from item_mast

where pro_price = (select MIN(pro_price) from item_mast);
```

	PRO_NAME	PRO_PRICE
1	Zip Drive	250
2	Mouse	250

Query 12 : Display all the customers, who are either belongs to the city New York or not had a grade above 100.

select * from customer

where customer_city = 'New York' or grade <= 100;

	CUSTOMER_ID	CUSTOMER_NAME	CUSTOMER_CITY	GRADE	SALESMAN_ID
1	3002	Nick Rimando	New York	100	5001
2	3007	Brad Davis	New York	200	5001
3	3009	Geoff Cameo	Berlin	100	(null)

Query 13 : Find those salesmen with all information who gets the commission within a range of 0.12 and 0.14.

select * from salesman

where commission between 0.12 and 0.14;

	SALESMAN_ID	NAME	CITY	COMMISSION
1	5002	Nail Knite	Paris	0.13
2	5006	Mc Lyon	Paris	0.14
3	5003	Lauson Hen	(null)	0.12
4	5007	Paul Adam	Rome	0.13

Query 14 : Find all those customers with all information whose names are ending with the letter 'n'.

Select * from customer

where customer_name like '%n';

	CUSTOMER_ID	CUSTOMER_NAME	CUSTOMER_CITY	GRADE	SALESMAN_ID
1	3001	Brad Guzan	London	(null)	(null)
2	3008	Julian Green	London	300	5002

Query 15 : Find those salesmen with all information whose name containing the 1st character is 'N' and the 4th character is 'l' and rests may be any character.

Select * from salesman

where Name like 'N__l%';

	SALESMAN_ID	NAME	CITY	COMMISSION
1	5002	Nail Knite	Paris	0.13

Query 16: Find that customer with all information who does not get any grade except NULL.

Select * from customer where grade is null;

	CUSTOMER_ID	CUSTOMER_NAME	CUSTOMER_CITY	GRADE	SALESMAN_ID
1	3001	Brad Guzan	London	(null)	(null)

Query 17 : Find the total purchase amount of all orders.

select sum(punch_amt) as Total_Amt from orders;

	TOTAL_AMT
1	14495.25

Query 18 : Find the number of salesman currently listing for all of their customers.

select count (distinct salesman_id) as Total_Salesman from orders

where salesman_id is not null;

	TOTAL_SALESMAN
1	4

Query 19 : Find the highest grade for each of the cities of the customers.

select customer_city, max(grade) as grade from customer

group by customer_city;

	CUSTOMER_CITY	GRADE
1	London	300
2	Paris	300
3	California	200
4	New York	200
5	Moncow	200
6	Berlin	100

Query 20 : Find the highest purchase amount ordered by the each customer with their ID and highest purchase amount.

```
select customer_id, max(punch_amt) as Max_Amt from orders

where customer_id is not null

group by customer_id

order by Max_Amt desc;
```

	CUSTOMER_ID	MAX_AMT
1	3002	5760
2	3009	2480
3	3007	2401
4	3004	1983
5	3005	948.5
6	3001	270.7
7	3008	250.5
8	3003	75.29

Query 21 : Find the highest purchase amount ordered by the each customer on a particular date with their ID, order date and highest purchase amount.

```
select customer_id, order_date, max(punch_amt) as Max_Amt from orders

where customer_id is not null

group by customer_id, order_date

order by Max_Amt desc;
```

	CUSTOMER_ID	ORDER_DATE	MAX_AMT
1	3002	10-09-16	5760
2	3009	10-10-16	2480
3	3007	27-07-16	2401
4	3004	10-10-16	1983
5	3005	10-09-16	948.5
6	3001	10-09-16	270.7
7	3008	27-06-16	250.5
8	3005	05-10-16	150.5
9	3009	17-08-16	110.5
10	3003	17-08-16	75.29
11	3002	05-10-16	65.26

Query 22 : Find the highest purchase amount on a date '2016-08-17' for each salesman with their ID.

Select salesman_id, max(punch_amt) as Max_Amt from orders

Where order_date = ’17-08-16’

Group by salesman_id

Order by Max_Amt desc;

	SALESMAN_ID	MAX_AMT
1	(null)	110.5
2	5007	75.29

Query 23 : Find the highest purchase amount with their customer ID and order date, for only those customers who have the highest purchase amount in a day is more than 2000.

select customer_id, order_date, max(punch_amt) as Max_Amt from orders

group by customer_id, order_date

having max(punch_amt) > 2000

order by Max_Amt;

	CUSTOMER_ID	ORDER_DATE	MAX_AMT
1	3007	27-07-16	2401
2	3009	10-10-16	2480
3	3002	10-09-16	5760

Query 24 : Write a SQL statement that counts all ordersfor a date August 17th, 2016.

select count(*) as Total_ORD_17_Aug from orders

where order_date = '17-08-16';

	TOTAL_ORD_17_AUG
1	2

Query 25 : Find the name and city of those customers and salesmen who lives in the same city.

select s.name as Salesman_Name, c.customer_name, c.customer_city from salesman s , customer c

where s.city = c.customer_city;

	SALESMAN_NAME	CUSTOMER_NAME	CUSTOMER_CITY
1	James Hoog	Nick Rimando	New York
2	Pit Alex	Brad Guzan	London
3	Nail Knite	Fabian Johns	Paris
4	Mc Lyon	Fabian Johns	Paris
5	James Hoog	Brad Davis	New York
6	Pit Alex	Julian Green	London

Query 26 : Find the names of all customers along with the salesmen who works for them.

Select s.name as salesman_name, c.customer_name from salesman s, customer c

Where s.salesman_id = c.salesman_id;

	SALESMAN_NAME	CUSTOMER_NAME
1	James Hoog	Nick Rimando
2	Nail Knite	Graham Zusi
3	Mc Lyon	Fabian Johns
4	James Hoog	Brad Davis
5	Nail Knite	Julian Green
6	Paul Adam	Jozy Altidor

Query 27 : Display all those orders by the customers not located in the same cities where their salesmen live.

Select o.order_no, c.customer_name, c.customer_city, s.name as salesman_name, s.city as salesman_city from orders o, customer c, salesman s

Where c.customer_city <> s.city

And o.customer_id = c.customer_id

And o.salesman_id = s.salesman_id;

	ORDER_NO	CUSTOMER_NAME	CUSTOMER_CITY	SALESMAN_NAME	SALESMAN_CITY
1	7001	Graham Zusi	California	Nail Knite	Paris
2	7007	Graham Zusi	California	Nail Knite	Paris
3	7012	Julian Green	London	Nail Knite	Paris
4	7011	Jozy Altidor	Moncow	Paul Adam	Rome

Query 28 : Display all the orders issued by the salesman 'Paul Adam' from the orders table.

```
Select o.order_no , s.salesman_id, s.name salesman_name from orders o, salesman s

Where s.name = 'Paul Adam'

And o.salesman_id = s.salesman_id;
```

	ORDER_NO	SALESMAN_ID	SALESMAN_NAME
1	7011	5007	Paul Adam

```
select * from orders

where salesman_id in (select salesman_id from salesman

                      where name = 'Paul Adam');
```

	ORDER_NO	PUNCH_AMT	ORDER_DATE	CUSTOMER_ID	SALESMAN_ID
1	7011	75.29	17-08-16	3003	5007

Query 29 : Display all the orders which values are greater than the average order value for 10th October 2012.

```
select * from orders where punch_amt >

  (select avg(punch_amt) from orders

      where order_date = '10-10-16');
```

	ORDER_NO	PUNCH_AMT	ORDER_DATE	CUSTOMER_ID	SALESMAN_ID
1	7005	2401	27-07-16	3007	5001
2	7008	5760	10-09-16	3002	5001
3	7003	2480	10-10-16	3009	(null)

Query 30 : Find all orders attributed to salesmen in Paris.

```
select * from orders where salesman_id

in (select salesman_id from salesman where city = 'Paris');
```

	ORDER_NO	PUNCH_AMT	ORDER_DATE	CUSTOMER_ID	SALESMAN_ID
1	7001	150.5	05-10-16	3005	5002
2	7007	948.5	10-09-16	3005	5002
3	7010	1983	10-10-16	3004	5006
4	7012	250.5	27-06-16	3008	5002

Query 31 : Extract the data from the orders table for the salesman who earned the maximum commission.

```
select * from orders where salesman_id
in (select salesman_id from salesman where commission =
(select max(commission) from salesman))
order by punch_amt desc;
```

	ORDER_NO	PUNCH_AMT	ORDER_DATE	CUSTOMER_ID	SALESMAN_ID
1	7008	5760	10-09-16	3002	5001
2	7005	2401	27-07-16	3007	5001
3	7002	65.26	05-10-16	3002	5001

Query 32 : Find the name and ids of all salesmen who had more than one customer.

```
select salesman_id, name from salesman s
where(select count(*) from customer c
                where c.salesman_id = s.salesman_id) > 1;
```

	SALESMAN_ID	NAME
1	5001	James Hoog
2	5002	Nail Knite

```
select c.salesman_id, s.name from customer c, salesman s
where c.salesman_id = s.salesman_id
group by c.salesman_id, s.name
having count(c.salesman_id)>1;
```

	SALESMAN_ID	NAME
1	5001	James Hoog
2	5002	Nail Knite

Query 33 : Write a query to find all the salesmen who worked for only one customer.

```
select c.salesman_id, s.name from customer c, salesman s
where c.salesman_id = s.salesman_id
group by c.salesman_id, s.name
having count(c.salesman_id)=1;
```

	SALESMAN_ID	NAME
1	5007	Paul Adam
2	5006	Mc Lyon

```
Select * from salesman
where salesman_id IN ( select distinct  salesman_id from customer a
                       where not exists ( select * from customer b
                                          where a.salesman_id = b.salesman_id
                                          and a.customer_name <> b.customer_name));
```

	SALESMAN_ID	NAME	CITY	COMMISSION
1	5006	Mc Lyon	Paris	0.14
2	5007	Paul Adam	Rome	0.13

Query 34 : Display all the orders that had amounts that were greater than at least one of the orders from September 10th 2016.

```
Select * from orders
where punch_amt > any
(select punch_amt from orders where order_date = '10-09-16');
```

	ORDER_NO	PUNCH_AMT	ORDER_DATE	CUSTOMER_ID	SALESMAN_ID
1	7008	5760	10-09-16	3002	5001
2	7003	2480	10-10-16	3009	(null)
3	7005	2401	27-07-16	3007	5001
4	7010	1983	10-10-16	3004	5006
5	7007	948.5	10-09-16	3005	5002

Query 35 : Display only those customers whose grade are, in fact, higher than every customer in New York.

```
select * from customer where grade >
all (select grade from customer where customer_city = 'New York');
```

	CUSTOMER_ID	CUSTOMER_NAME	CUSTOMER_CITY	GRADE	SALESMAN_ID
1	3004	Fabian Johns	Paris	300	5006
2	3008	Julian Green	London	300	5002

```
select * from customer where grade >
(select max(grade) from customer where customer_city = 'New York');
```

	CUSTOMER_ID	CUSTOMER_NAME	CUSTOMER_CITY	GRADE	SALESMAN_ID
1	3004	Fabian Johns	Paris	300	5006
2	3008	Julian Green	London	300	5002